CSC1042 Computer Network II

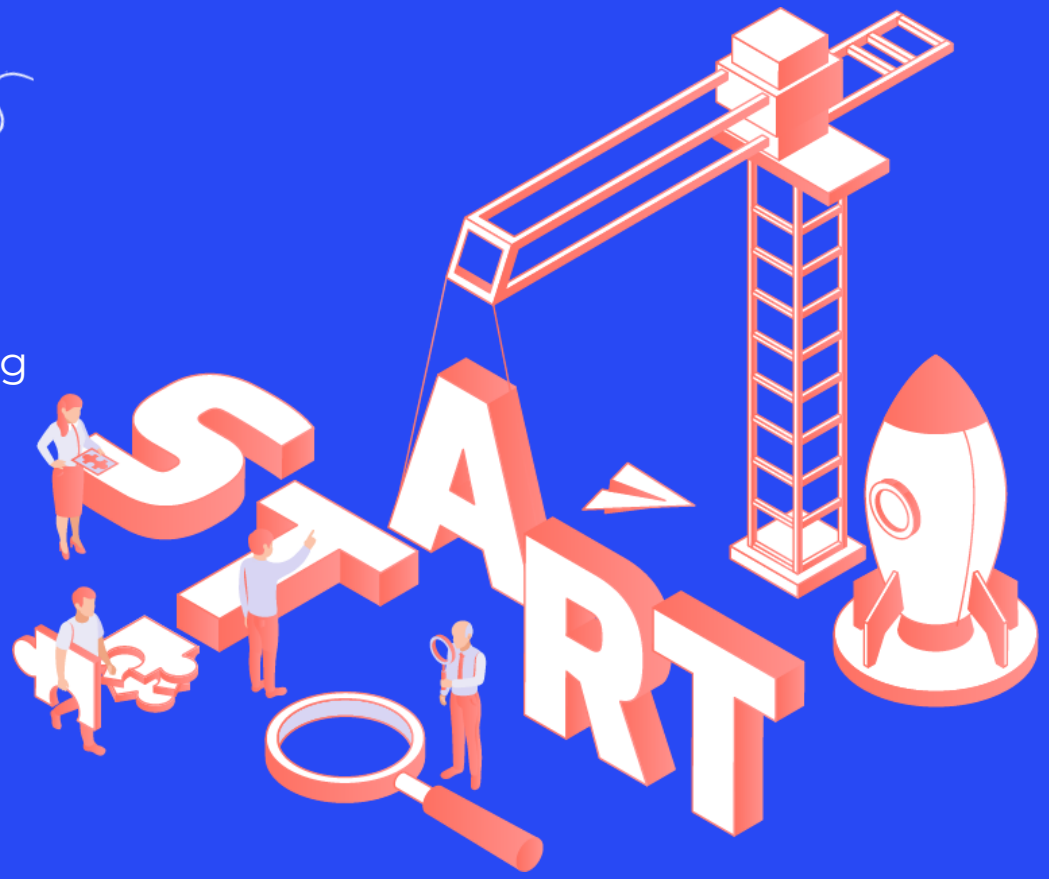
Link State Routing

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Learning outcomes

- Explain the concept of link state routing
- Describe how routers build and update their routing tables using Link State routing
- Describe how OSPF protocol works



Link state routing

- Distance vector routing took long to converge → we needed something better
- Link state routing was proposed
- It maintains more information about the internetwork than distance vector routing
  - Apart from the routing tables, link state routers maintain two additional tables
    - Neighbour table: contains the routers that are available to exchange routing data with
    - Topology table: contains all destination networks and every neighbour through which it can be reached. Also called Link State Database (LSDB)

Link state routing – How it works?

- Neighbour discovery
  - On booting, router A sends out Hello message on every link it is connected to
  - Routers on the other end of the links should respond by identifying themselves
    - This include IP address, subnet mask, etc.
  - Router A records this information into its *neighbour table*

Link state routing – How it works?

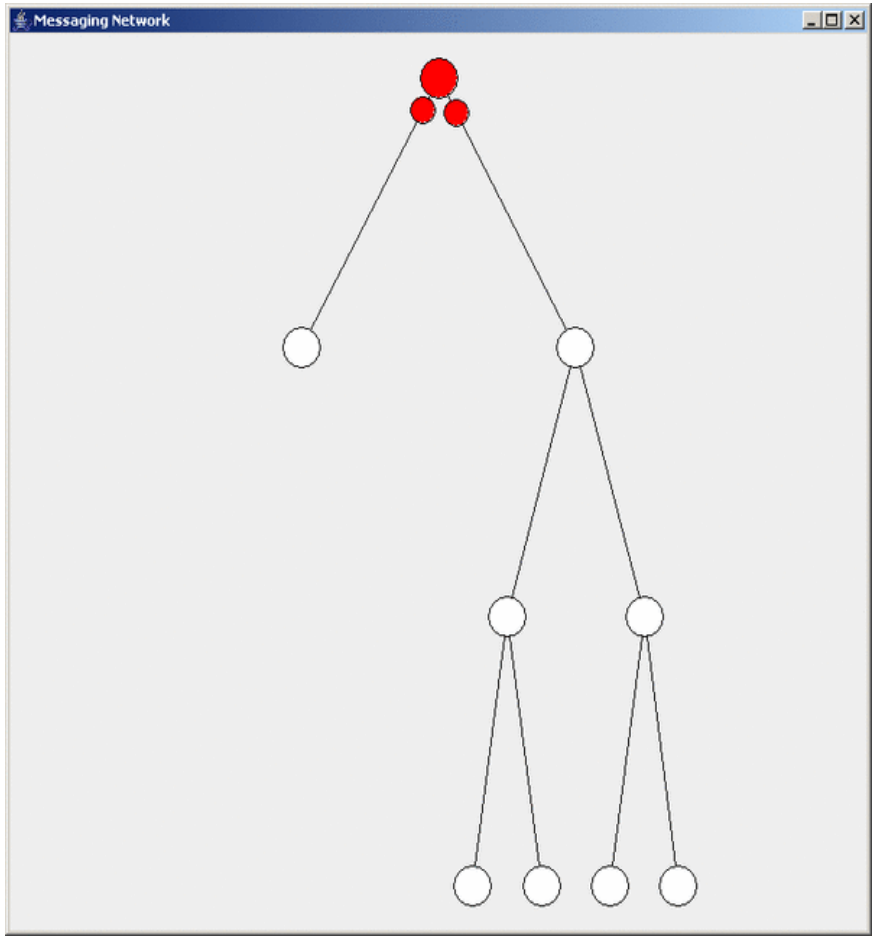
- Measuring link distance
  - Link distance can be computed based on delay, bandwidth, or a combination of the two
  - Routers use special echo packets to bounce off other routers and time the transit
    - If the traffic load is to be considered, the timer is started when the echo packet is queued
    - Otherwise, the timer is started once the echo packet is about to be transmitted

Link state routing – How does it work?

- Building a link state advertisement (LSA)
  - Each router builds up an LSA containing its neighbours and their corresponding distance
  - The LSA starts with the ID of the sending router
  - SEQ no. and Age fields follow
    - The age field is used to ensure that no packet will wander around indefinitely (equivalent to TTL)
  - When to build them?
    - Periodically or if a change in topology happened

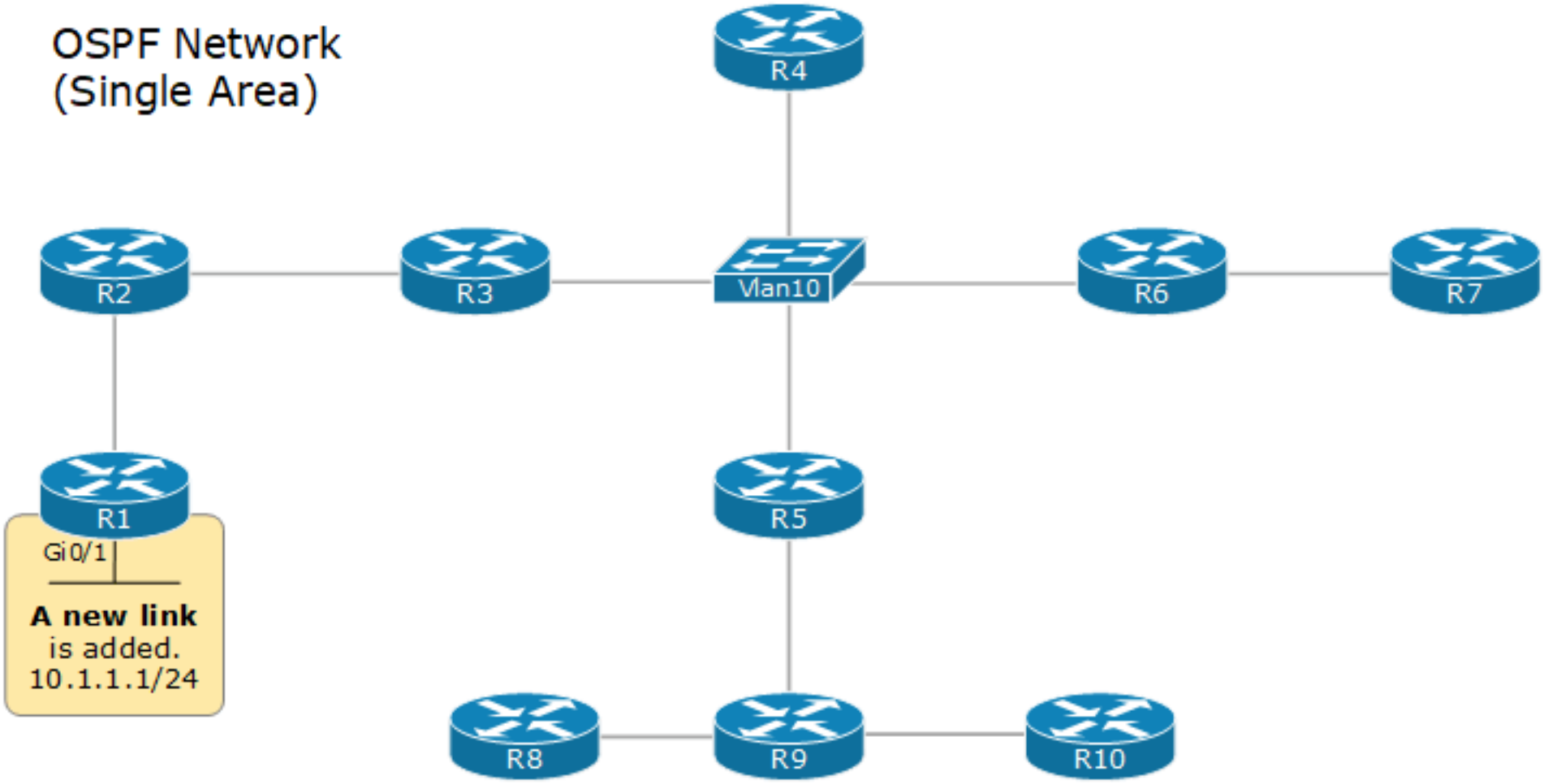
Link state routing – How does it work?

- Distributing LSAs
  - Use flooding to distribute LSAs
    - SEQ no. constitute a version field (i.e., only new versions are considered)
    - Records for the current SEQ no. for routers are maintained
  - On receiving a new LSA, routers check SEQ no.
    - If new, they'll flood it
    - Otherwise, they'll drop it



https://commons.wikimedia.org/wiki/File:Flooding_routing.gif

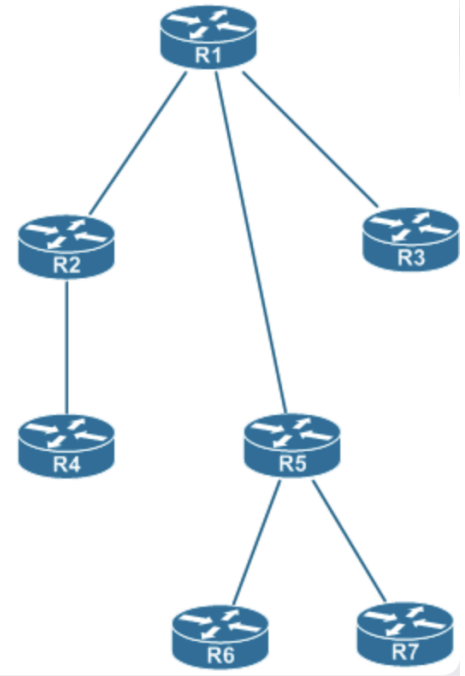
Link state routing – How does it work?



https://www.networkacademy.io/ccna/ospf/how-does-ospf-work

Link state routing – How does it work?

- Constructing topology and computing routes
  - When receiving LSAs, routers construct the network topology by building the topology table (LSDB)
  - Once the topology is completed, routers use Dijkstra's algorithm to compute the shortest path to all remote destination networks
  - Some challenges
    - Large networks can entail large RAM at the routers
    - Router failures may give rise to incorrect routing information



Open Shortest Path First (OSPF)

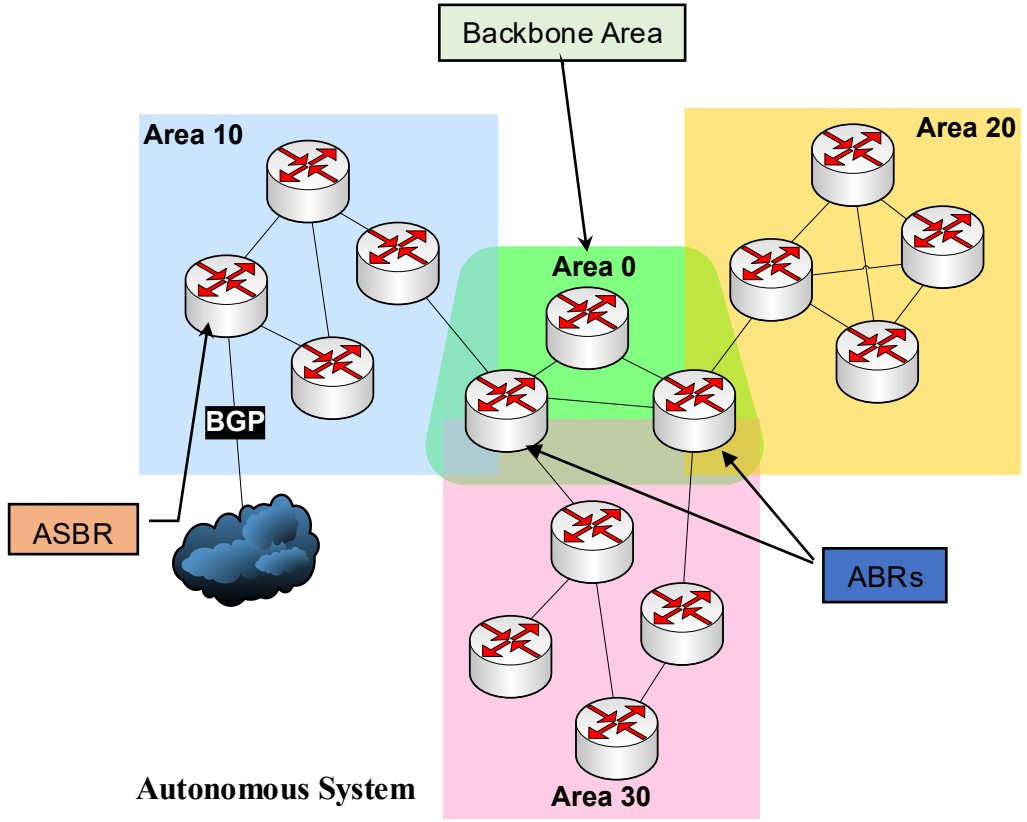
- A Link State routing protocol
- It's an interior gateway protocol
- Very wildly used
- It is a Layer 3 protocol of the OSI model
  - Protocol ID 89
- Route updates are sent only when a link changes state (e.g., fails)
- Routers use multicast to send route updates

Open Shortest Path First (OSPF)

- OSPF introduces the concept of “areas”
  - It divides the AS into smaller parts
  - This is to speed up the shortest path computations
- Every AS has a backbone area (area 0) and several other areas
- Routers in each area know only the topology of their respective area
- Some routers can be members of more than one area; they are called Area Border Routers (ABRs)
  - They forward routing information between areas
- Autonomous System Border Routers (ASBRs) connect AS to the Internet

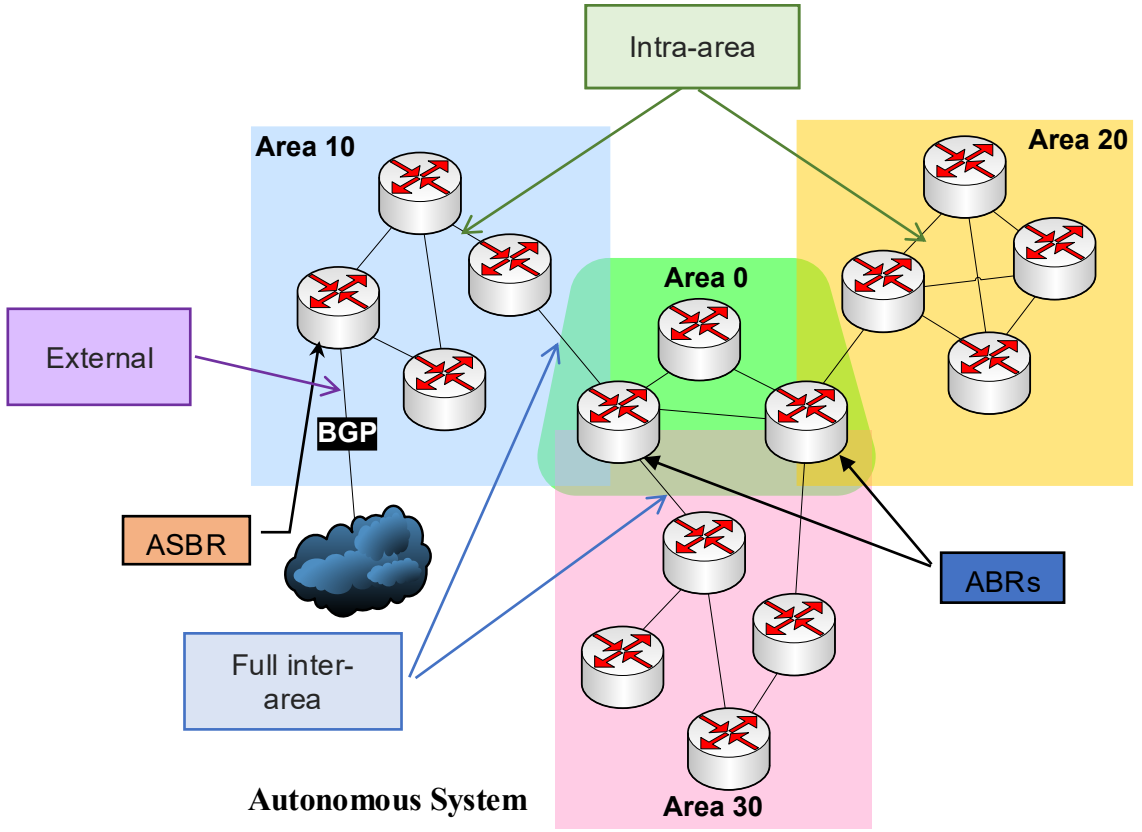
Open Shortest Path First (OSPF)

- All routers in the backbone area should be directly connected
- Routers in other areas can be indirectly connected
- All areas should be connected to the backbone area
  - All inter-area communications must pass through the backbone area
  - There is no direct link between two non-backbone areas



OSPF – route types

- OSPF defines three types of routes
  - Intra-area: routes within the same area
  - Full inter-area: routes between different areas
  - External: routes to the Internet

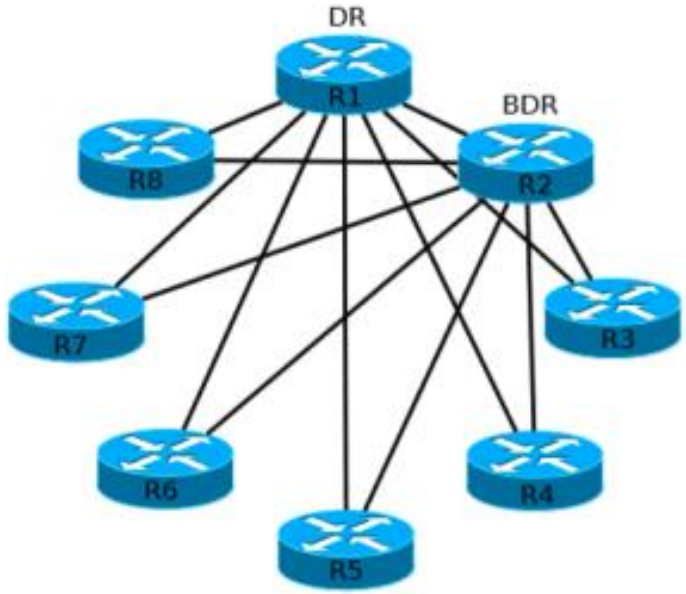


OSPF – area types

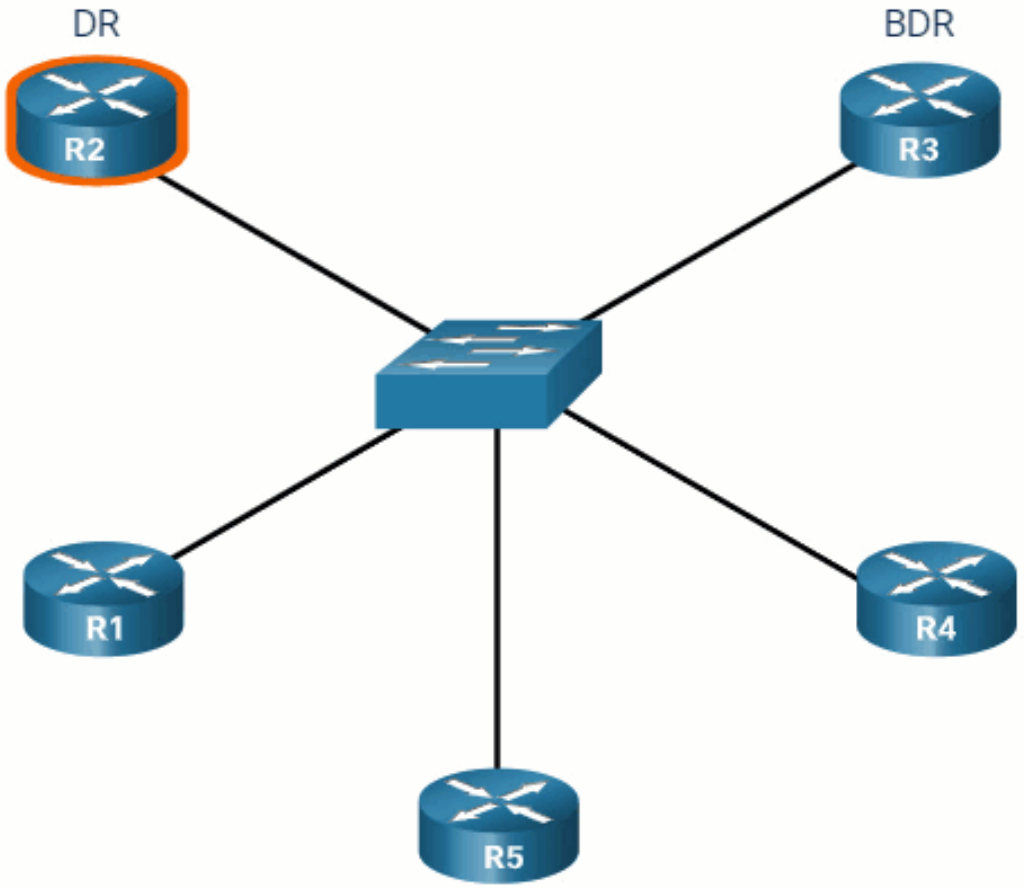
- Stub
  - Area without ASBR (e.g., areas 20 and 30)
  - External routing information not advertised within this area. Instead, it is replaced by a default route
- Totally Stubby
  - Neither external routing information nor routing information from other areas are advertised
- Not-So-Stubby
  - Stub area that contains ASBR
  - External routing information learned by routers in this area can be advertised to other areas
  - External routing information learned by routers from outside this area is not advertised within the area

OSPF – designated router

- For large networks, sharing the routing information with all the routers in the network is an issue
  - High routing overhead
- Routers in every area can vote to make one router a Designated Router (DR)
- Routers send and receive routing information to and from the designated router only
- Routers also select a Backup Designated Router (BDR) in case DR fails



OSPF – DR and BDR



https://itexamanswers.net/ccna-3-v7-0-curriculum-module-2-single-area-ospfv2-configuration.html

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